SQL Advance Analytics Project

1. Introduction

This project focuses on applying advanced analytical techniques to transactional sales data in order to move beyond basic exploration and uncover trends, performance drivers, and actionable business insights. Using SQL-based analytics, the project analyzes how the business evolves over time, how different products and customers perform relative to benchmarks, and how revenue is distributed across categories and segments. The outcome is a structured, insight-driven analysis that supports strategic decision-making and reporting.

2. Business Background

The business operates in a multi-product retail environment with diverse customers, varying price ranges, and sales activity spread across multiple years. While the data was previously explored through exploratory data analysis (EDA), stakeholders now require deeper analytical insights to understand growth patterns, identify top and underperforming products, evaluate customer value, and support performance monitoring. This Advanced Analytics project addresses those needs by transforming historical sales data into meaningful metrics, segments, and reporting views that can be used by analysts, managers, and BI tools for ongoing business evaluation.

3. Tools Used

To complete this project, I used the following tools:

- SQL – Core tool for data exploration, joins, aggregations, and analysis
- PostgreSQL – Relational database used to store and query the data
- pgAdmin / VS Code – For writing and executing SQL queries

4. Data Model Overview

The database follows a star schema, commonly used in analytics systems.

Tables Used:

fact_sales

- Contains transactional sales data
- Metrics: quantity sold, revenue, discounts, order date

dim_customers

- Customer attributes such as segment, country, and region

dim_products

- Product details including category, price, and product name

Relationships:

- fact_sales.customer_key → dim_customers.customer_key
- fact_sales.product_key → dim_products.product_key

This structure enables complex joins, aggregations, and performance analysis across multiple dimensions.

5. The Analysis

Each SQL query in this project was designed to answer a specific business question.

1. Change-Over-Time Trends Analysis

Business Question

How have sales, customer activity, and order volume evolved over time, and are there seasonal patterns that should influence planning, forecasting, and resource allocation?

Approach

Aggregated sales, customers, and quantity sold at:

- Year level → to identify long-term growth patterns
- Month level → to uncover seasonality

Used order_date as the time dimension and analyzed only valid sales records
Focused on business-level metrics, not transaction noise

SQL Queries Used

```sql
-- Analyze Sales Performance Over Time (Yearly).
SELECT
  EXTRACT(YEAR FROM order_date) AS order_year,
  SUM(sales_amount) AS total_sales,
  COUNT(DISTINCT customer_key) AS total_customers,
  SUM(quantity) AS total_quantity
FROM
  fact_sales
```

```sql
WHERE
  order_date IS NOT NULL
GROUP BY
  EXTRACT(YEAR FROM order_date)
ORDER BY
  EXTRACT(YEAR FROM order_date)
;
```

order_year	total_sales	total_customers	total_quantity
Filter...	Filter...	Filter...	Filter...
2010	43419	14	14
2011	7075088	2216	2216
2012	5842231	3255	3397
2013	16344878	17427	52807
2014	45642	834	1970

```sql
-- Analyze Sales Performance Over Time (Monthly).
SELECT
  EXTRACT(MONTH FROM order_date) AS order_month,
  SUM(sales_amount) AS total_sales,
  COUNT(DISTINCT customer_key) AS total_customers,
  SUM(quantity) AS total_quantity
FROM
  fact_sales
WHERE
  order_date IS NOT NULL
GROUP BY
  EXTRACT(MONTH FROM order_date)
ORDER BY
  EXTRACT(MONTH FROM order_date)
;
```

order_month	total_sales	total_customers	total_quantity
Filter...	Filter...	Filter...	Filter...
1	1868558	1818	4043
2	1744517	1765	3858
3	1908375	1982	4449
4	1948226	1916	4355
5	2204969	2074	4781
6	2935883	2430	5573
7	2412838	2154	5107
8	2684313	2312	5335
9	2536520	2210	5070
10	2916550	2533	5838
11	2979113	2500	5756
12	3211396	2656	6239

Key Outcomes & Insights

1. Year-Over-Year Sales Trends (Strategic View)

Year	Total Sales	Customers	Quantity Sold
2010	43K	14	14
2011	7.1M	2,216	2,216
2012	5.8M	3,255	3,397
2013	**16.3M**	**17,427**	**52,807**
2014	46K	834	1,970

Insights:

- 2013 is the breakout year, contributing the majority of total revenue and volume
- Customer growth peaked sharply in 2013, indicating:
  - Successful market expansion or major product push
- 2010 and 2014 show partial or incomplete data, likely representing:
  - Initial launch (2010)
  - Data cutoff or incomplete year (2014)

Business Implication:
Strategic decisions (forecasting, hiring, inventory) should be anchored to 2011–2013 trends, not edge years with incomplete data.

2. Monthly Sales Trends (Seasonality View)

Month	Total Sales	Customers	Quantity
Jan	1.87M	1,818	4,043
Feb	1.74M	1,765	3,858
Mar	1.91M	1,982	4,449
Apr	1.95M	1,916	4,355
May	2.20M	2,074	4,781
Jun	2.94M	2,430	5,573
Jul	2.41M	2,154	5,107
Aug	2.68M	2,312	5,335
Sep	2.54M	2,210	5,070
Oct	2.92M	2,533	5,838
Nov	2.98M	2,500	5,756
Dec	**3.21M**	**2,656**	**6,239**

Insights:

- Strong upward momentum from May onward
- Q4 (Oct–Dec) is the highest-performing period:
    - December is the peak month across all metrics
- Early months (Jan–Feb) consistently underperform

Business Implication:

- Marketing spends, promotions, and inventory planning should be front-loaded for Q4
- Q1 is ideal for:
    - Cost control
    - Testing initiatives
    - Infrastructure optimization

Why This Analysis Matters

This time-based analysis enables stakeholders to:

- Separate true growth from seasonal spikes
- Build accurate forecasts
- Align staffing, inventory, and marketing budgets with demand cycles

2. Cumulative Analysis

Business Question

Is the business demonstrating sustained growth over time, and how do sales accumulation patterns differ across months and years? Additionally, how is pricing evolving as the business scales?

Approach

Calculated monthly total sales and applied:

- Running totals across the full timeline → to assess long-term growth
- Yearly running totals → to compare intra-year momentum

Computed monthly average price and tracked a moving average to smooth volatility and identify pricing trends

Used window functions (**SUM() OVER**, **AVG() OVER**) to simulate real-world executive reporting metrics

SQL Queries Used

```sql
-- Calculate the total sales per month and the running total of sales over time.
SELECT
  order_date,
  total_sales,
  SUM(total_sales) OVER(ORDER BY order_date ASC) AS running_total
FROM (
  SELECT
    DATE_TRUNC('month', order_date) AS order_date,
    SUM(sales_amount) AS total_sales
  FROM
    fact_sales
  WHERE
    order_date IS NOT NULL
  GROUP BY
    DATE_TRUNC('month', order_date)
) t
;
```

order_date	total_sales	running_total
Filter...	Filter...	Filter...
2010-12-01 00:00:00+05	43419	43419
2011-01-01 00:00:00+05	469795	513214
2011-02-01 00:00:00+05	466307	979521
2011-03-01 00:00:00+05	485165	1464686
2011-04-01 00:00:00+05	502042	1966728
2011-05-01 00:00:00+05	561647	2528375
2011-06-01 00:00:00+05	737793	3266168
2011-07-01 00:00:00+05	596710	3862878
2011-08-01 00:00:00+05	614516	4477394
2011-09-01 00:00:00+05	603047	5080441
2011-10-01 00:00:00+05	708164	5788605
2011-11-01 00:00:00+05	660507	6449112
2011-12-01 00:00:00+05	669395	7118507
2012-01-01 00:00:00+05	495363	7613870

```sql
-- Calculate the total sales per month and the running total of sales over
time but on yearly basis.
SELECT
  order_date,
  total_sales,
  SUM(total_sales) OVER(PARTITION BY DATE_TRUNC('year', order_date) ORDER BY
order_date ASC) AS running_total
FROM (
  SELECT
    DATE_TRUNC('month', order_date) AS order_date,
    SUM(sales_amount) AS total_sales
  FROM
    fact_sales
  WHERE
    order_date IS NOT NULL
  GROUP BY
    DATE_TRUNC('month', order_date)
) t
;
```

order_date	total_sales	running_total
Filter...	Filter...	Filter...
2010-12-01 00:00:00+05	43419	43419
2011-01-01 00:00:00+05	469795	469795
2011-02-01 00:00:00+05	466307	936102
2011-03-01 00:00:00+05	485165	1421267
2011-04-01 00:00:00+05	502042	1923309
2011-05-01 00:00:00+05	561647	2484956
2011-06-01 00:00:00+05	737793	3222749
2011-07-01 00:00:00+05	596710	3819459
2011-08-01 00:00:00+05	614516	4433975
2011-09-01 00:00:00+05	603047	5037022
2011-10-01 00:00:00+05	708164	5745186
2011-11-01 00:00:00+05	660507	6405693
2011-12-01 00:00:00+05	669395	7075088
2012-01-01 00:00:00+05	495363	495363

```sql
-- Calculate the average price per month and the moving average price over
time.
SELECT
  order_date,
  avg_price,
  ROUND(AVG(avg_price) OVER(ORDER BY order_date ASC), 2) AS
moving_average_price
FROM (
  SELECT
    DATE_TRUNC('month', order_date) AS order_date,
    ROUND(AVG(price), 2) AS avg_price
  FROM
    fact_sales
  WHERE
    order_date IS NOT NULL
  GROUP BY
    DATE_TRUNC('month', order_date)
) t
;
```

order_date	avg_price	moving_average...
Filter...	Filter...	Filter...
2010-12-01 00:00:00+05	3101.36	3101.36
2011-01-01 00:00:00+05	3262.47	3181.92
2011-02-01 00:00:00+05	3238.24	3200.69
2011-03-01 00:00:00+05	3234.43	3209.13
2011-04-01 00:00:00+05	3197.72	3206.84
2011-05-01 00:00:00+05	3227.86	3210.35
2011-06-01 00:00:00+05	3207.80	3209.98
2011-07-01 00:00:00+05	3173.99	3205.48
2011-08-01 00:00:00+05	3184.02	3203.10
2011-09-01 00:00:00+05	3259.71	3208.76
2011-10-01 00:00:00+05	3204.36	3208.36
2011-11-01 00:00:00+05	3175.51	3205.62
2011-12-01 00:00:00+05	3015.29	3190.98
2012-01-01 00:00:00+05	1965.73	3103.46

Key Outcomes & Insights

1. Cumulative Sales Growth (Overall Timeline)

- Cumulative revenue increased steadily from $43K in Dec 2010 to approximately $29.35M by Jan 2014
- Growth accelerated significantly after 2011, with:
    - 2011 ending at $7.1M
    - 2012 closing at $12.96M cumulative
    - 2013 alone contributing over $16.3M, more than 55% of total lifetime revenue

Insight:

- The business exhibits strong compounding growth, with momentum heavily concentrated in later years—an indicator of successful scaling rather than stagnant performance.

2. Yearly Cumulative Patterns (Intra-Year Momentum)

- 2011: Gradual, consistent accumulation ending at $7.07M
- 2012: Slower mid-year growth but recovered strongly in Q4, closing at $5.84M
- 2013: Rapid acceleration:
    - Crossed $5M by May
    - Surpassed $10M by September
    - Closed the year at $16.34M, the strongest annual performance

Insight:

- 2013 shows a much steeper accumulation curve, signaling improved demand, stronger customer acquisition, or expanded product reach compared to prior years.

3. Average Price & Moving Average Trends

- Average monthly price declined sharply over time:
  - ~$3,200 in early 2011
  - ~$1,700 by late 2012
  - ~$300–350 throughout 2013
- The moving average price dropped from $3,101 to $1,746 by early 2014

Insight:

Revenue growth is increasingly driven by higher volume rather than higher prices, suggesting:

- A shift toward lower-priced products
- Expanded mass-market penetration
- Possible strategic pricing or discounting initiatives

Business Implications

The business is scaling successfully, with cumulative sales confirming long-term growth
2013 represents a structural shift, not a temporary spike
Declining prices paired with rising revenue indicate:

- Improved accessibility
- Higher transaction volume
- A volume-led growth strategy

3. Performance Analysis

Business Question

How did each product perform over time compared to its historical average sales and previous year's performance, and which products showed consistent growth, decline, or volatility?

Approach

Aggregated yearly product-level sales from the fact table.
Used window functions to:

- Calculate each product's average annual sales.
- Compare current-year sales against that average.
- Perform Year-over-Year (YoY) comparisons using lag analysis.

Classified performance into:

- Above Average / Below Average

- Increase / Decrease vs Previous Year

SQL Queries Used

```
-- Analyze the yearly performance of products by comparing each product's
sale to both its average sales performance and the previous year's sales.

WITH yearly_product_sales AS(
  SELECT
    EXTRACT(YEAR FROM order_date) AS order_year,
    p.product_name AS product_name,
    SUM(f.sales_amount) AS current_sales
  FROM
    fact_sales AS f
  LEFT JOIN
    dim_products AS p
    ON f.product_key = p.product_key
  WHERE
    f.order_date IS NOT NULL
  GROUP BY
    EXTRACT(YEAR FROM order_date),
    p.product_name
)
SELECT
  order_year,
  product_name,
  current_sales,
  ROUND(AVG(current_sales) OVER(PARTITION BY product_name), 2) AS avg_sales,
  current_sales - ROUND(AVG(current_sales) OVER(PARTITION BY product_name),
2) AS diff_avg,
  CASE
    WHEN current_sales - AVG(current_sales) OVER(PARTITION BY product_name)
> 0 THEN 'Above Avg'
    WHEN current_sales - AVG(current_sales) OVER(PARTITION BY product_name)
< 0 THEN 'Below Avg'
    ELSE 'Avg'
  END AS avg_change,

  LAG(current_sales) OVER(PARTITION BY product_name ORDER BY order_year) AS
pre_yr_sales,
  current_sales - LAG(current_sales) OVER(PARTITION BY product_name ORDER BY
order_year) AS diff_py_sales,

  CASE
    WHEN current_sales - LAG(current_sales) OVER(PARTITION BY product_name
ORDER BY order_year) > 0 THEN 'Increase'
    WHEN current_sales - LAG(current_sales) OVER(PARTITION BY product_name
ORDER BY order_year) < 0 THEN 'Decrease'
    ELSE 'No Change'
  END AS py_change
FROM
```

```
  yearly_product_sales
ORDER BY
  product_name,
  order_year
;
```

order_year	product_name	current_sales	avg_sales	diff_avg	avg_change	pre_yr_sales	diff_py_sales	py_change
Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...
2012	All-Purpose Bike Stand	159	13197.00	-13038.00	Below Avg	NULL	NULL	No Change
2013	All-Purpose Bike Stand	37683	13197.00	24486.00	Above Avg	159	37524	Increase
2014	All-Purpose Bike Stand	1749	13197.00	-11448.00	Below Avg	37683	-35934	Decrease
2012	AWC Logo Cap	72	6570.00	-6498.00	Below Avg	NULL	NULL	No Change
2013	AWC Logo Cap	18891	6570.00	12321.00	Above Avg	72	18819	Increase
2014	AWC Logo Cap	747	6570.00	-5823.00	Below Avg	18891	-18144	Decrease
2013	Bike Wash - Dissolver	6960	3636.00	3324.00	Above Avg	NULL	NULL	No Change
2014	Bike Wash - Dissolver	312	3636.00	-3324.00	Below Avg	6960	-6648	Decrease
2013	Classic Vest- L	11968	6240.00	5728.00	Above Avg	NULL	NULL	No Change
2014	Classic Vest- L	512	6240.00	-5728.00	Below Avg	11968	-11456	Decrease
2013	Classic Vest- M	11840	6368.00	5472.00	Above Avg	NULL	NULL	No Change
2014	Classic Vest- M	896	6368.00	-5472.00	Below Avg	11840	-10944	Decrease
2012	Classic Vest- S	64	3648.00	-3584.00	Below Avg	NULL	NULL	No Change
2013	Classic Vest- S	10368	3648.00	6720.00	Above Avg	64	10304	Increase
2014	Classic Vest- S	512	3648.00	-3136.00	Below Avg	10368	-9856	Decrease

Key Outcomes and Insights

2013 emerged as the strongest performance year across most products.

- Many products significantly exceeded their historical averages.
- Example: All-Purpose Bike Stand achieved 37,683 in sales, outperforming its average by +24,486.

2014 showed a sharp decline across nearly all products.

- Several products fell well below both their historical averages and prior-year sales.
- Example: AWC Logo Cap dropped from 18,891 (2013) to 747 (2014).

High Year-over-Year volatility was observed:

- Strong upward momentum from 2012 → 2013, followed by steep reversals in 2013 → 2014.
- This pattern was consistent across both high-revenue and low-revenue products.

Business Implications

- Product performance is highly period-dependent, making simple averages misleading without trend context.
- Significant YoY fluctuations indicate potential external factors (market shifts, seasonality, or partial-year data).
- This analysis enables:
    - Early identification of overperforming products
    - Detection of declining or unstable SKUs

- More informed inventory, pricing, and forecasting decisions

4. Part-to-Whole Analysis

Business Question

Which product categories contribute the most to overall sales, and how concentrated is revenue across categories?

Approach

- Aggregated total sales by product category.
- Calculated overall sales using a window function.
- Computed each category's percentage contribution to total revenue.
- Ranked categories by total sales to identify dominant revenue drivers.

SQL Queries Used

```sql
-- Which categories contribute the most to overall sales?
WITH cte_category_sales AS (
  SELECT
    p.category AS category,
    SUM(f.sales_amount) AS total_sales
  FROM
    fact_sales AS f
  LEFT JOIN
    dim_products AS p
    ON f.product_key = p.product_key
  GROUP BY
    p.category
)
SELECT
  category,
  total_sales,
  SUM(total_sales) OVER() AS overall_sales,
  CONCAT(ROUND((total_sales / SUM(total_sales) OVER()) * 100, 2), '%') AS
percentage_of_total
FROM
  cte_category_sales
ORDER BY
  total_sales DESC
;
```

category	total_sales	overall_sales	percentage_of_t...
Filter...	Filter...	Filter...	Filter...
Bikes	28316272	29356250	96.46%
Accessories	700262	29356250	2.39%
Clothing	339716	29356250	1.16%

Key Outcomes and Insights

- Bikes overwhelmingly dominate total revenue, contributing 96.46% of overall sales.
- Accessories account for 2.39%, while Clothing contributes only 1.16%.
- Revenue distribution is highly concentrated, with non-bike categories playing a marginal role.

Business Implications

The business is heavily dependent on a single category (Bikes), increasing risk exposure. Accessories and Clothing present growth and diversification opportunities if strategically expanded.
Insights support:

- Portfolio diversification decisions
- Category-level investment prioritization
- Bundling and cross-selling strategies to increase non-bike revenue

5. Data Segmentation

Business Question

How are products and customers distributed across meaningful segments, and what does this reveal about pricing structure and customer value?

Approach

Product Segmentation

- Classified products into cost-based ranges using defined price thresholds.
- Counted products in each cost segment to understand pricing distribution.

Customer Segmentation

- Calculated each customer's:
  - Total lifetime spending
  - Relationship lifespan (in months)
- Segmented customers into:
  - VIP: ≥12 months lifespan and spending > $5,000

    - Regular: ≥12 months lifespan and spending ≤ $5,000
    - New: lifespan < 12 months
- Aggregated customer counts by segment.

SQL Queries Used

```sql
-- Segment products into cost ranges and count how many products fall into
each segment.
WITH cte_product_segments AS (
  SELECT
    product_key,
    product_name,
    cost,
    CASE
      WHEN cost < 100 THEN 'Below 100'
      WHEN cost BETWEEN 100 AND 500 THEN '100-500'
      WHEN cost BETWEEN 500 AND 1000 THEN '500-1000'
      ELSE 'Above 1000'
    END AS cost_range
  FROM
    dim_products
)
SELECT
  cost_range,
  COUNT(product_key) AS total_products
FROM
  cte_product_segments
GROUP BY
  cost_range
ORDER BY
  cost_range
;
```

cost_range	total_products
Filter...	Filter...
100-500	101
500-1000	45
Above 1000	39
Below 100	110

```sql
-- Group customers into three segments based on their spending behavior:
  -- VIP: at least 12 months of history and spending more than $5000.
  -- Regular: at least 12 months of history but spending $5000 or less.
  -- New: lifespan less than 12 months.
-- And find the total number of customers by each group.
```

```sql
WITH cte_customer_spending AS (
  SELECT
    c.customer_key,
    SUM(f.sales_amount) AS total_spending,
    MIN(order_date) AS first_order,
    MAX(order_date) AS last_order,
    EXTRACT(YEAR FROM AGE(MAX(order_date), MIN(order_date))) * 12 +
EXTRACT(MONTH FROM AGE(MAX(order_date), MIN(order_date))) AS lifespan
  FROM
    fact_sales AS f
  LEFT JOIN
    dim_customers AS c
    ON f.customer_key = c.customer_key
  GROUP BY
    c.customer_key
),
cte_customer_segments AS(
  SELECT
    customer_key,
    total_spending,
    lifespan,
    CASE
      WHEN lifespan >= 12 AND total_spending > 5000 THEN 'VIP'
      WHEN lifespan >= 12 AND total_spending <= 5000 THEN 'Regular'
      ELSE 'NEW'
    END AS customer_segment
  FROM
    cte_customer_spending
)
SELECT
  customer_segment,
  COUNT(customer_key) AS total_customers
FROM
  cte_customer_segments
GROUP BY
  customer_segment
ORDER BY
  total_customers DESC
;
```

customer_segment	total_customers
Filter...	Filter...
NEW	14828
Regular	2037
VIP	1619

Key Outcomes and Insights

Product Segmentation

- The largest product groups fall in the Below $100 (110 products) and $100–$500 (101 products) ranges.
- Fewer products exist in higher-cost tiers:
  - $500–$1000: 45 products
  - Above $1000: 39 products
- Indicates a catalog skewed toward low to mid-priced offerings.

Customer Segmentation

- New customers dominate the base (14,828 customers), indicating strong acquisition.
- Regular customers total 2,037, while VIP customers account for 1,619.
- A relatively small VIP group likely contributes a disproportionate share of revenue.

Business Implications

Product pricing structure supports high-volume, lower-cost sales, with room to expand premium offerings.
Strong inflow of new customers highlights effective acquisition but signals a retention and upsell opportunity.
Clear segmentation enables:

- Targeted loyalty programs for VIPs
- Conversion strategies for New → Regular customers
- Personalized marketing and pricing optimization

6. Reporting – Customer Analytics Report

Business Question

How can we create a single, reusable customer report that captures customer demographics, purchasing behavior, value segmentation, and actionable KPIs for downstream analytics and reporting?

Approach

This analysis builds a customer-level analytical view designed for reporting and BI consumption.

1. Base Data Assembly

- Joined sales and customer dimensions to capture:
  - Transaction details (orders, sales, quantity)
  - Customer attributes (name, age)
- Filtered to valid transactions only.

2. Customer-Level Aggregation

- Aggregated metrics per customer:
  - Total orders
  - Total sales
  - Total quantity purchased
  - Unique products purchased
  - Customer lifespan (in months)
  - Last order date

3. Segmentation & Enrichment

- Age Groups: Under 20, 20–29, 30–39, 40–49, 50+
- Customer Segments:
  - VIP: ≥12 months lifespan & > $5,000 spend
  - Regular: ≥12 months lifespan & ≤ $5,000 spend
  - New: < 12 months lifespan

4. KPI Calculations

- Recency: Months since last purchase
- Average Order Value (AOV): Total sales ÷ total orders
- Average Monthly Spend: Total sales ÷ lifespan

5. Data Modeling for Reuse

- Final query structured as a database view to enable:
  - Power BI dashboards
  - Self-service analytics
  - Consistent metric definitions across teams

SQL Queries Use

```
CREATE VIEW report_customers AS
(
WITH cte_base_query AS (
-- 1) Base Query: Retrieves core columns from tables
  SELECT
    f.order_number,
    f.product_key,
    f.order_date,
    f.sales_amount,
    f.quantity,
    c.customer_key,
    c.customer_number,
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    DATE_PART('year', AGE(NOW(), c.birthdate)) AS age
  FROM
    fact_sales AS f
  LEFT JOIN
```

```
    dim_customers AS c
    ON f.customer_key = c.customer_key
  WHERE
    order_date IS NOT NULL
), cte_customer_aggregation AS (
-- 2) Customer Aggregations: Summarizes key metrics at the customer level
  SELECT
    customer_key,
    customer_number,
    customer_name,
    age,
    COUNT(DISTINCT order_number) AS total_orders,
    SUM(sales_amount) AS total_sales,
    SUM(quantity) AS total_quantity,
    COUNT(DISTINCT product_key) AS total_products,
    MAX(order_date) AS last_order_date,
    EXTRACT(YEAR FROM AGE(MAX(order_date), MIN(order_date))) * 12 +
EXTRACT(MONTH FROM AGE(MAX(order_date), MIN(order_date))) AS lifespan
  FROM
    cte_base_query
  GROUP BY
    customer_key,
    customer_number,
    customer_name,
    age
)
SELECT
  customer_key,
  customer_number,
  customer_name,
  age,

  CASE
    WHEN age < 20 THEN 'Under 20'
    WHEN age BETWEEN 20 AND 29 THEN '20-29'
    WHEN age BETWEEN 30 AND 39 THEN '30-39'
    WHEN age BETWEEN 40 AND 49 THEN '40-49'
    ELSE '50 and above'
  END AS age_group,

  CASE
    WHEN lifespan >= 12 AND total_sales > 5000 THEN 'VIP'
    WHEN lifespan >= 12 AND total_sales <= 5000 THEN 'Regular'
    ELSE 'NEW'
  END AS customer_segment,

  total_orders,
  total_sales,
  total_quantity,
  total_products,
  last_order_date,
  EXTRACT(YEAR FROM AGE(NOW(), last_order_date)) * 12 + EXTRACT(MONTH FROM
AGE(NOW(), last_order_date)) AS recency,
```

```
  -- Compute average Order Value
  CASE
    WHEN total_sales = 0 THEN 0
    WHEN total_orders = 0 THEN 0
    ELSE total_sales / total_orders
  END AS avg_order_value,

  -- Compute average monthly spend
  CASE
    WHEN lifespan = 0 THEN total_sales
    ELSE ROUND(total_sales / lifespan, 2)
  END AS avg_monthly_spend
FROM
  cte_customer_aggregation
)
;
```

c	custome...	custome...	age	age_gro...	custo...	total_ord...	total_sales	t.	t.	last_ord...	rece...	avg_o...	avg...
1	AW00011000	Jon Yang	54	50 and above	VIP	3	8249	8	8	2013-05-03	152	2749	305.52
2	AW00011001	Eugene Huang	49	40-49	VIP	3	6384	11	10	2013-12-10	145	2128	187.76
3	AW00011002	Ruben Torres	54	50 and above	VIP	3	8114	4	4	2013-02-23	154	2704	324.56
4	AW00011003	Christy Zhu	52	50 and above	VIP	3	8139	9	9	2013-05-10	152	2713	290.68
5	AW00011004	Elizabeth John...	46	40-49	VIP	3	8196	6	6	2013-05-01	152	2732	303.56
6	AW00011005	Julio Ruiz	49	40-49	VIP	3	8121	6	6	2013-05-02	152	2707	290.04
7	AW00011006	Janet Alvarez	49	40-49	VIP	3	8119	5	5	2013-05-14	152	2706	300.70
8	AW00011007	Marco Mehta	56	50 and above	VIP	3	8211	8	8	2013-03-19	153	2737	315.81
9	AW00011008	Rob Verhoff	50	50 and above	VIP	3	8106	7	7	2013-03-02	154	2702	324.24
10	AW00011009	Shannon Carls...	56	50 and above	VIP	3	8091	5	5	2013-05-09	152	2697	299.67
11	AW00011010	Jacquelyn Sua...	56	50 and above	VIP	3	8088	4	4	2013-05-23	151	2696	288.86
12	AW00011011	Curtis Lu	56	50 and above	VIP	3	8133	4	4	2013-03-19	153	2711	312.81

Key Outcomes and Insights

Each customer is represented by a single, analytics-ready record.

Enables simultaneous analysis of:

- Customer value
- Engagement frequency
- Product diversity
- Lifecycle stage

Clear separation between high-value (VIP), long-term (Regular), and early-stage (New) customers.

Recency and spend metrics allow identification of:

- At-risk customers
- High-potential upsell candidates
- Loyal, high-frequency buyers

Business Implications

Creates a scalable reporting foundation for marketing, sales, and leadership teams.
Supports:

- Targeted retention campaigns
- Personalized promotions
- Customer lifetime value analysis

Eliminates repeated metric logic by centralizing calculations in a trusted analytical view.
Directly enables BI tools (e.g., Power BI) to focus on insight delivery, not data preparation.

6. Conclusion

Key Insights

This Advanced Analytics project transformed raw transactional data into actionable business intelligence by applying multiple analytical perspectives to the same dataset.

- Change-over-time analysis revealed clear growth patterns and seasonality, with strong sales momentum peaking toward year-end and a major revenue expansion phase in 2013.
- Cumulative analysis confirmed sustained long-term growth, while year-partitioned running totals highlighted performance consistency and volatility across different years.
- Performance analysis identified products that consistently outperform their historical averages and flagged underperformers through year-over-year comparisons.
- Part-to-whole analysis showed that the business is heavily driven by a single category, with Bikes contributing over 96% of total revenue, indicating both strength and concentration risk.
- Data segmentation uncovered meaningful customer and product groupings, distinguishing high-value VIP customers from Regular and New segments, and classifying products across clear cost tiers.
- Reporting views consolidated customer behavior, value, and engagement metrics into a reusable analytical layer ready for BI tools and executive dashboards.

Together, these analyses provide a 360-degree view of performance, growth, and customer value.

Closing Thoughts

This project demonstrates how advanced SQL analytics can move beyond descriptive reporting into decision-ready insights. By layering trends, cumulative metrics, performance benchmarks, segmentation, and reporting structures, the analysis enables businesses to:

- Monitor growth and seasonality with confidence
- Evaluate product and category performance objectively
- Identify high-value customers and retention risks
- Build scalable reporting pipelines for ongoing analysis

The final output is not just analysis, but a strategic analytics foundation that can directly support forecasting, performance management, and data-driven decision-making across the organization.